BRAC UNIVERSITY
Inspiring Excellence

CSE422: Artificial intelligence

Quiz 04 [Summer 2026]

Time: **20 minutes**, Full Marks: **15**

A

Name:		**ID:**		**Section:**	

1. Consider a linear regression model with the following data samples [you can extend the table on the right if needed]:

x_1	x_2	y	y'	$y'-y$	$(y'-y)^2$	$(y'-y)\cdot x_1$	$(y'-y)\cdot x_2$
1	2	5	-1	-6	36	-6	-12
2	-1	1	3	2	4	4	-2
0	3	6	-3	-9	81	0	-27
					$\Sigma = 121$	$\Sigma = -2$	$\Sigma = -41$

The model uses a modified mean square error loss function $L = \frac{1}{4N}\sum_{i=1}^{N}(Y' - y)^2$; where N is the number of data samples. **Initial parameters: w1 = 1, w2 = –1, b = 0; learning rate = 0.01.**

(a) What should be the linear regression equation for this model? [1]
(b) Find out the equations for the gradients of the loss function with respect to w1 and w2. [4]
(c) Calculate the total loss. [4]
(d) Apply one iteration of gradient descent to find the updated values of w1 and w2. [6]

(a) $y' = w_1x_1 + w_2x_2 + b$

(b) $L = \frac{1}{4N}\Sigma(y'-y)^2 = \frac{1}{4N}\Sigma(w_1x_1 + w_2x_2 + b - y)^2$

$\frac{\partial L}{\partial w_1} = \frac{1}{4N}\cdot 2\cdot\Sigma(w_1x_1+w_2x_2+b-y)\cdot x_1 = \frac{1}{2N}\Sigma(y'-y)\cdot x_1$

$\frac{\partial L}{\partial w_2} = \frac{1}{2N}\Sigma(y'-y)\cdot x_2$

(c) Here, $y' = 1\cdot x_1 + (-1)\cdot x_2 + 0 = x_1 - x_2$; calculations in table.

$L = \frac{1}{4\times 3}\times 121 = \frac{121}{12} = \boxed{10.083}$

(d) $\frac{\partial L}{\partial w_1} = \frac{1}{2\times 3}\times(-2) = -\frac{1}{3}$; $\frac{\partial L}{\partial w_2} = \frac{1}{2\times 3}\times(-41) = -\frac{41}{6}$

$w_1' = w_1 - \alpha\cdot\frac{\partial L}{\partial w_1'} \rightarrow w_1 = 1 - 0.01\times\left(-\frac{1}{3}\right) = \boxed{1.0033}$

$w_2 = (-1) - 0.01\times\left(-\frac{41}{6}\right) = \boxed{-0.9317}$

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B

Name:		**ID:**		**Section:**	

1. Consider a linear regression model with the following data samples [you can extend the table on the right if needed]:

x_1	x_2	y	y'	$y'-y$	$(y'-y)^2$	$(y'-y)\cdot x_1$	$(y'-y)\cdot x_2$
1	2	5	1	−4	16	−4	−8
0	-2	1	−2	−3	9	0	6
2	4	6	2	−4	16	−8	−16
					$\Sigma = 41$	$\Sigma = -12$	$\Sigma = -18$

The model uses a modified mean square error loss function $L = \frac{1}{6N}\sum_{i=1}^{N}(Y' - y)^2$; where N is the number of data samples. **Initial parameters: w1 = −1, w2 = 1, b = 0; learning rate = 0.01.**

(a) What should be the linear regression equation for this model? [1]
(b) Find out the equations for the gradients of the loss function with respect to w1 and w2. [4]
(c) Calculate the total loss. [4]
(d) Apply one iteration of gradient descent to find the updated values of w1 and w2. [6]

(a) $y' = w_1x_1 + w_2x_2 + b$

(b) $L = \frac{1}{6N}\Sigma(y'-y)^2 = \frac{1}{6N}\Sigma(w_1x_1 + w_2x_2 + b - y)^2$

$\frac{\partial L}{\partial w_1} = 2\cdot\frac{1}{6N}\cdot\Sigma(w_1x_1 + w_2x_2 + b - y)\cdot x_1 = \frac{1}{3N}\Sigma(y'-y)\cdot x_1$

$\frac{\partial L}{\partial w_2} = \frac{1}{3N}\Sigma(y'-y)\cdot x_2$

(c) Here, $y' = (-1)x_1 + 1\cdot x_2 + 0 = -x_1 + x_2$

$L = \frac{1}{6\times3}\times 41 = \frac{41}{18} = \boxed{2.278}$

(d) $\frac{\partial L}{\partial w_1} = \frac{1}{3\times6}\times(-12) = -\frac{2}{3}$; $\frac{\partial L}{\partial w_2} = \frac{1}{3\times6}\times(-18) = -1$

$w_1 = (-1) - 0.01\times\left(-\frac{2}{3}\right) = \boxed{-0.9933}$

$w_2 = 1 - 0.01\times(-1) = \boxed{1.01}$